Aditya Potnis

Email: apotnis2@illinois.edu
LinkedIn: linkedin.com/in/adityapotnis/
Mobile: 774-636-0696
Github: github.com/adipotnis Website: adipotnis.github.io
Champaign, IL

Summary

CS & Robotics graduate student with 4.5+ years experience in robot learning, motion planning and control systems for autonomous vehicles and robots. Proficient in C++, Python, ROS2. Passionate about building safe, intelligent technology at scale.

Education

- **University of Illinois, Urbana-Champaign** — **Urbana, IL**
  Master of Science - Computer Science. GPA: 4.0/4.0 — May 2026
  Courses: Humanoid Robotics, Autonomous Vehicle System Engineering, Deep Learning for Computer Vision, 3D Vision, Artificial Intelligence, Networked Internet of Things, Software Engineering

- **College of Engineering Pune** — **Pune, India**
  Bachelor of Technology - Mechanical Engineering. CGPA: 8.01/10.0 — Aug 2018 - Jun 2022
  Courses: Mechanical Measurement and Automatic Control, Robotics and Automation, Engineering Mechanics, Linear Algebra, Calculus, Object Oriented Programming & Data Structures and Algorithms (Minor in Computer Engineering).

Publications

CTS-MoE: Implicit Policy Adaptation Enables Perceptive Locomotion on Discontinuous Terrain — Under Review: *CoRL 2026*
Francisco Affonso, Matheus P. Angarola, Ana Luiza Mineiro, Aditya Potnis, Marcelo Becker, Girish Chowdhary

CATNAV: Cached Vision-Language Traversability for Efficient Zero-Shot Robot Navigation — Under Review
Aditya Potnis, Francisco Affonso, Shreya Gummadi, Naveen Kumar Uppalapati, Girish Chowdhary

Visual-Language-Guided Task Planning for Horticultural Robots — *Computers and Electronics in Agriculture*
Jose Cuaran, Kendall Koe, Aditya Potnis, Naveen Kumar Uppalapati, Girish Chowdhary

Skills

Programming: C++, Python, MATLAB, ROS, ROS2 , Bash, PyTorch, Tensorflow, MS Excel

Tools & CAD Software: Gazebo, IsaacSim, Git, Docker, Linux, Fusion360, SolidWorks, Rapid Prototyping

Core Areas: Reinforcement Learning, Robot Learning, Embodiment Reasoning, Motion Planning, SLAM, 3D Vision

Experience

Woven by Toyota — **Palo Alto, CA**
Software Engineering Intern, Motion planning — May 2025 - Aug 2025

- Identified performance metrics and developed analysis tool to compare Rapidly-Exploring Random Tree (rule-based) and Data Driven Planning methods for Trajectory Optimization Warm Start with optimal safety and comfort constraints.

Distributed Autonomous Systems (DAS) Lab, UIUC — **Urbana, IL**
Research Assistant — Aug 2024 - Present

- Training Vision Language Action Models (VLAs) for motion planning and generalized multimodal robotic navigation, for motion primitives such as jumping, walking, and non-holonomic steering.
- Prototyped heuristics for Sampling-based motion planning algorithms (RRT, RRT*) to navigate Octree based occupancy and affordance map generated by VLM-assisted behavior-driven semantic segmentation using CLIPSeg and utilized MPPI for control.
- Implemented Zero-Shot Affordance estimation and SLAM using OpenVINS for the Salto-1P Jumping Robot.
- Research supported by NASA FireSense, AIFARMS, I-FARM, COALESCE, and NSF ACCESS.

Earthsense Inc. — **Champaign, IL**
Robotics Engineer — Ju1 2022 - Aug 2024

- Owned the low-level control stack deployed across 200+ TerraSentia field robots and a 750kg-payload UGV with Dual-Ackermann steering and 4-wheel torque vectoring over the Peak CAN bus, improving runtime performance with 80% fewer interventions.
- Migrated and optimized the MPPI planning & control stack to ROS2, improving inference throughput and applying QoS tuning to cut communication bandwidth by 30%.
- Developed software-in-loop and hardware-in-loop simulation pipelines in Gazebo using Docker and behavior trees.
- Led field testing across India and Malaysia; applied FMEA and Design-for-Manufacturing principles to co-design robot steering with the hardware team.

Earthsense Inc. **Pune, India**
Robotics and Autonomy System Development Intern Jun 2021 - Aug 2021

- Developed error diagnostic suite using ROS bag data to plot robot waypoints, navigation statistics & collision detection using Python, OpenCV and Matplotlib, reducing batch processing by 3 times (15 minutes to 5 minutes).
- Implemented automated robot performance monitoring pipeline using docker and AWS to keep track of Key Performance Indicators, reducing overall workflow analysis time by 20%.

Teaching Experience

University of Illinois, Urbana-Champaign **Urbana, IL**
Teaching Assistant, CS 427 Software Engineering I Jan 2026 - May 2026

- Built automated grading tooling and developed course projects on mini SWE coding agents, exposing students to LLM-driven autonomous code generation and review workflows.
- Graded team project deliverables against software engineering rubrics (design, documentation, testing), and mentored students on object-oriented design and agile development practices.

Projects

TrajReasoner: Vision–Language Trajectory Reasoning with a GRPO Verifier Feb 2026 - May 2026

- Fine-tuning a vision–language model as a trajectory reasoner with GRPO, using verifiable rewards over traversability and proprioception to align generated motion plans with feasibility and goal-reaching constraints.

Perceptive Locomotion in Humanoid Robots via Mixture of Experts Oct 2025 - Dec 2025

- Worked on a Mixture of Experts based multi task RL training for humanoids using Proximal Policy Optimization. Implemented IsaacLab + RSL_RL simulation with 16% better stability metrics than baseline.

Low-Rank Adaptation for Video Generation with descriptive relative pose Oct 2025 - Nov 2025

- Fine-tuned Wan 2.2 (1.3B) with LoRA to generate robot-POV navigation videos from text + motion plans (e.g., "move forward 1.6 m, turn 14° right near a brick wall") for scalable outdoor data synthesis.
- Collected a multimodal outdoor robot dataset (RGB, GPS+IMU EKF poses, motion captions) and evaluated motion fidelity, identifying memory artifacts and noise sensitivity.

Visual Navigation Policy with V-JEPA2 and Diffusion Policy Oct 2025 - Dec 2025

- Trained a visual navigation policy using a frozen V-JEPA2 backbone for spatiotemporal feature extraction from egocentric video, with a Diffusion Policy action head for action prediction.

ADAPT: Diffusion-Predicted Pedestrian Avoidance with MPPI Control (CS 588) Feb 2026 - May 2026

- Built an autonomous driving stack on the Polaris GEM e4 pairing a diffusion-based pedestrian predictor with an MPPI planner for pedestrian-aware avoidance.
- Developed a text-promptable LiDAR-camera module for open-vocabulary goal specification.

GhibliDream – Studio-Ghibli Stylization of Stable Diffusion (CS 444) Jan 2025 - May 2025

- Fine-tuned StableDiffusion-2.0 for stylization with DreamBooth on curated images using a T4/A100 instance. Built a LLM-assisted (Gemini-2.0/GPT4o based) image auto-captioning workflow to generate variation of text length, tone and styles.
- Stress tested system via targeted ablation of samplers & prompt length. Reached 0.90+ CLIP-I cosine similarity on foreground characters while retaining backgrounds quality.

Simulating parabolic trajectory jumping robots Feb 2025 - Present

- Implemented and designed a parabolic jumping robot simulation using Gazebo to simulate Salto-1P as a point object to test indoor unstructured autonomous navigation using VLM based prompting (Gemini 2.0 and Qwen2.5).

Robotic serial arm manipulator using Bowden cables & Mechanical multiplexing Sep 2021 - May 2022

- Led the development of a compact, waterproof, 6-DOF Bowden cable-driven robotic arm with a novel mechanical multiplexer, enabling control of multi-jointed arms using only two stepper motors, reducing cost and size.
- Developed FK/IK control using C++ and ROS, structural analysis in ANSYS FEA and design in Fusion 360.

Honors and Awards

- Best Working Project Award (Mechanical Dept.), COEP; provisional patent filed (IN202321006687) Aug 2022
- Team India Coach, 2018 FIRST Lego League European Championships, Tallinn, Estonia — 2nd in Best Project Research, 4th in Overall Robot Game.
- Team Lead, Team India at World Robot Olympiad Internationals — Costa Rica 2017 (Top 15); India 2016 (Top 40), Gold Medal at 2016 Nationals.
- Team Lead, Top 10 at FIRST Lego League Open European Championship, Spain 2016 (sponsored by Tata Motors).

Volunteer Experience

- **Coach & Educator, Robominds**: Developed STEM curriculum for ages 5 to 15. Taught STEM robotics courses to 200+ children across Pune city. (Mar 2017 - Jun 2024)
- **Member, SIGRobotics (UIUC)**: Active member of the student robotics special interest group. (Nov 2025 - Present)

Work Authorization: U.S. Citizen